

Brian Acosta

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EXPERIENCE

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- Figure AI** 2025-Present
Senior Robotics/AI Engineer - Humanoid Robot Control San Jose, CA
- Research, development, deployment, and support of humanoid robot control systems
- UPenn GRASP Lab (General Robotics, Automation, Sensing & Perception Lab)** 2020-2025
Graduate Researcher Philadelphia, PA
- Built a perception and control stack in C++/Python for dynamic bipedal locomotion over rough terrain
 - Developed a real-time mixed-integer MPC footstep planner for bipedal walking over rough terrain
 - Developed a temporally stable terrain segmentation algorithm for vision-based reactive footstep planning
 - Successfully deployed novel perception and control algorithms on the real Cassie biped
- Intuitive Surgical** Summer 2023
Systems Analyst (Controls) Intern Sunnyvale, CA
- Developed and validated integration tests and analysis tools for minimally invasive surgical robot behaviors
- John Deere** Summer 2020
Controls Engineering Intern Dubuque IA (Remote due to COVID-19)
- Automated mass property validation of CAD models used for rigid body dynamics simulations
 - Designed and deployed a fan drive controller for a prototype construction vehicle

SELECT PUBLICATIONS

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- Brian Acosta** and Michael Posa. Perceptive Mixed-Integer Footstep Control for Underactuated Bipedal Walking on Rough Terrain. *IEEE Transactions on Robotics (T-RO)*, 2025.
 - Minku Kim, **Brian Acosta**, Pratik Chaudhary, and Michael Posa. Learning a Vision-Based Footstep Planner for Hierarchical Walking Control. *IEEE-RAS International Conference on Humanoid Robotics*, 2025.
 - Brian Acosta** and Michael Posa. Bipedal Walking on Constrained Footholds with MPC Footstep Control. *IEEE-RAS International Conference on Humanoid Robotics*, 2023.
 - Brian Acosta***, William Yang*, and Michael Posa. Validating Robotics Simulators On Real World Impacts. *IEEE Robotics and Automation Letters*, 2022.

TECHNICAL SKILLS

Programming: C++, Python, MATLAB/Simulink, C, Git, Linux

Robotics and Controls: Model-Predictive Control, Reinforcement learning, Debugging, Convex Optimization, Perception, Trajectory Optimization, Multibody Dynamics and Kinematics, Simulation

Tools and Frameworks: Drake, Eigen, PyTorch, OpenCV, PCL, Pybind11, LCM, ROS, Bazel

EDUCATION

University of Pennsylvania Philadelphia, PA
Ph.D. Mechanical Engineering and Applied Mechanics | NSF Graduate Research Fellowship May 2025

University of Pennsylvania Philadelphia, PA
M.S.E Robotics December 2024

Purdue University West Lafayette, IN
B.S. Mechanical Engineering | National Merit Scholarship, Purdue Presidential Scholarship May 2020